

Data Types and Missing Values

Dataset Shape: 489 rows x 63 columns

	Data Type	Missing Values	% Missing
case_id	object	0	0.000000
gender	object	0	0.000000
vital_status	object	0	0.000000
vital_days_after_surgery	int64	0	0.000000
bmi	float64	1	0.204499
age_at_nephrectomy	int64	0	0.000000
hospitalization	int64	0	0.000000
er_visit	object	2	0.408998
readmission	object	2	0.408998
surgery_type	object	0	0.000000
surgical_approach	object	1	0.204499
surgical_procedure	object	0	0.000000
estimated_blood_loss	float64	9	1.840491
clavien_surgical_complications	object	0	0.000000
ischemia_time	float64	230	47.034765
operative_time	float64	5	1.022495
smoking_history	object	0	0.000000
age_when_quit_smoking	float64	339	69.325153
pack_years	float64	236	48.261759
chewing_tobacco_use	object	1	0.204499
alcohol_use	object	1	0.204499
radiographic_size	float64	7	1.431493
cytoreductive	object	2	0.408998
pathologic_size	float64	21	4.294479
malignant	object	7	1.431493
pathology_t_stage	object	13	2.658487
pathology_n_stage	object	14	2.862986
pathology_m_stage	object	15	3.067485
tumor_histologic_subtype	object	6	1.226994
tumor_necrosis	object	13	2.658487
tumor_isup_grade	object	9	1.840491
positive_resection_margins	object	22	4.498978
sarcomatoid_features	object	13	2.658487
rhabdoid_features	object	13	2.658487
aua_risk_score	object	6	1.226994
comorbidities.myocardial_infarction	bool	0	0.000000
comorbidities.congestive_heart_failure	bool	0	0.000000
comorbidities.peripheral_vascular_disease	bool	0	0.000000
comorbidities.cerebrovascular_disease	object	1	0.204499
comorbidities.dementia	bool	0	0.000000
comorbidities.copd	bool	0	0.000000
comorbidities.connective_tissue_disease	bool	0	0.000000
comorbidities.peptic_ulcer_disease	bool	0	0.000000
comorbidities.uncomplicated_diabetes_mellitus	bool	0	0.000000
comorbidities.diabetes_mellitus_with_end_organ_damage	bool	0	0.000000
... (truncated for space)			

Summary Statistics (Numerical)

	count	mean	std	variance	min	max
vital_days_after_surgery	489.0	1034.460123	928.453842	862026.535804	0.0	7450.00
bmi	488.0	30.851926	6.968472	48.559600	16.1	72.23
age_at_nephrectomy	489.0	59.333333	14.392099	207.132514	0.0	90.00
hospitalization	489.0	3.345603	4.110311	16.894658	0.0	59.00
estimated_blood_loss	480.0	341.020833	588.405325	346220.826287	0.0	9000.00
ischemia_time	259.0	19.691120	11.305019	127.803448	0.0	59.00
operative_time	484.0	228.727273	95.411347	9103.325052	44.0	613.00
age_when_quit_smoking	150.0	42.720000	14.827473	219.853960	0.0	72.00
pack_years	253.0	13.082213	22.413476	502.363909	0.0	210.00
radiographic_size	482.0	4.998506	3.366209	11.331364	0.5	18.00
pathologic_size	468.0	5.082179	3.472805	12.060375	0.5	22.50
last_preop_egfr.days_before_nephrectomy	393.0	41.974555	221.224225	48940.157514	0.0	4057.00
last_preop_egfr.value	297.0	68.138047	14.319867	205.058581	5.0	90.00
first_postop_egfr.days_after_nephrectomy	397.0	334.103275	585.215679	342477.390823	0.0	3403.00
first_postop_egfr.value	337.0	57.706231	16.230475	263.428324	9.0	89.00
last_postop_egfr.days_after_nephrectomy	386.0	922.005181	792.103450	627427.875298	1.0	3715.00
last_postop_egfr.value	324.0	57.981481	17.553447	308.123495	5.0	90.00

Summary Statistics (Categorical)

Value counts for case_id:

case_id	
case_00000	1
case_00001	1
case_00002	1
case_00003	1
case_00004	1

Value counts for gender:

gender	
male	306
female	182
transgender_male_to_female	1

Value counts for vital_status:

vital_status	
censored	412
dead	77

Value counts for er_visit:

er_visit	
False	433
True	54
None	2

Value counts for readmission:

readmission	
False	438
True	49
None	2

Value counts for surgery_type:

surgery_type	
robotic	287
open	116
laparoscopic	65
percutaneous	21

Value counts for surgical_approach:

surgical_approach	
transperitoneal	424
retroperitoneal	61
trans_to_retro	3
None	1

Value counts for surgical_procedure:

surgical_procedure	
partial_nephrectomy	275
radical_nephrectomy	177
cryoablation	24
nephroureterectomy	13

Value counts for clavien_surgical_complications:

clavien_surgical_complications	
0	401
1	39
2	23
3b	10
3a	7

Value counts for smoking_history:

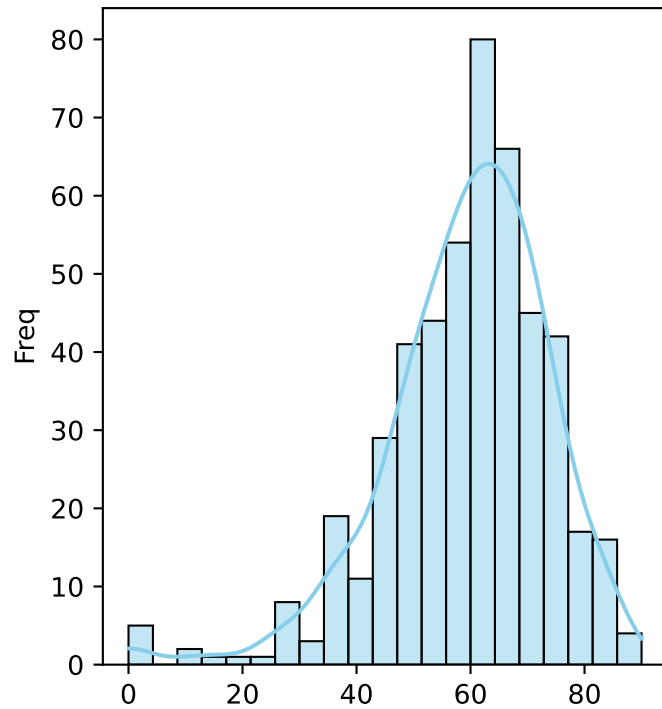
smoking_history	
never_smoked	224
previous_smoker	190
current_smoker	71
not_found_in_emr	4

Value counts for chewing_tobacco_use:

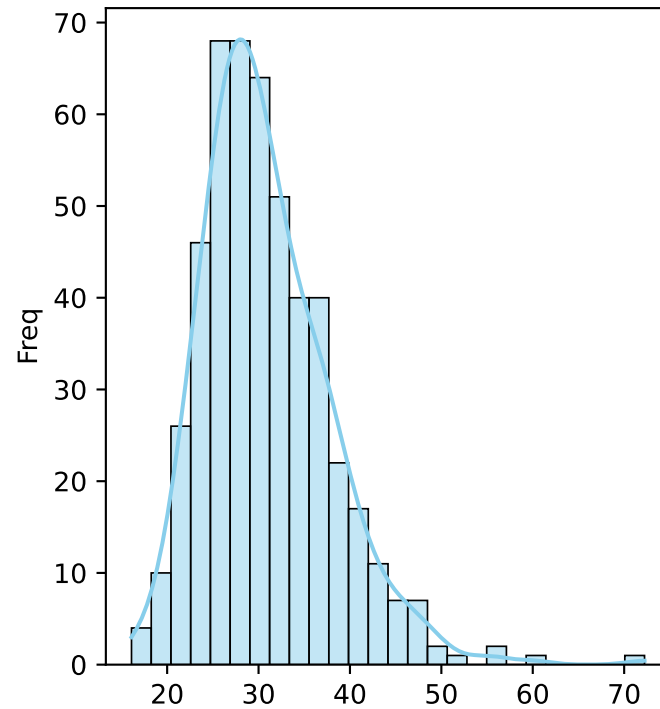
chewing_tobacco_use	
never or not in the last 3 months	476

Distributions of Key Numerical Variables

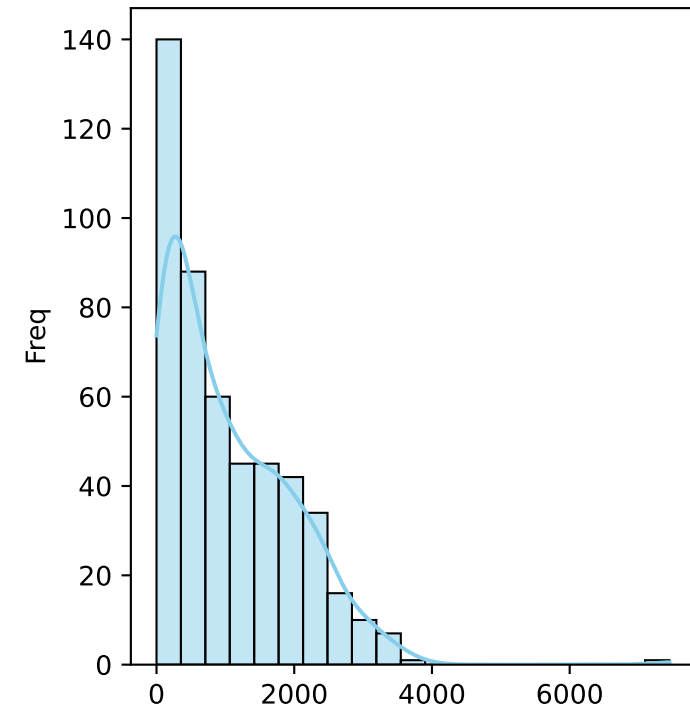
age_at_nephrectomy



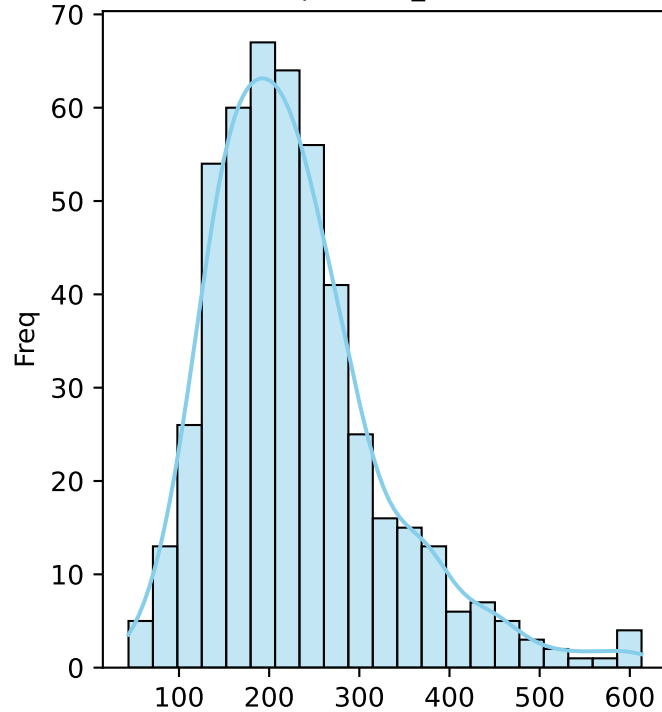
bmi



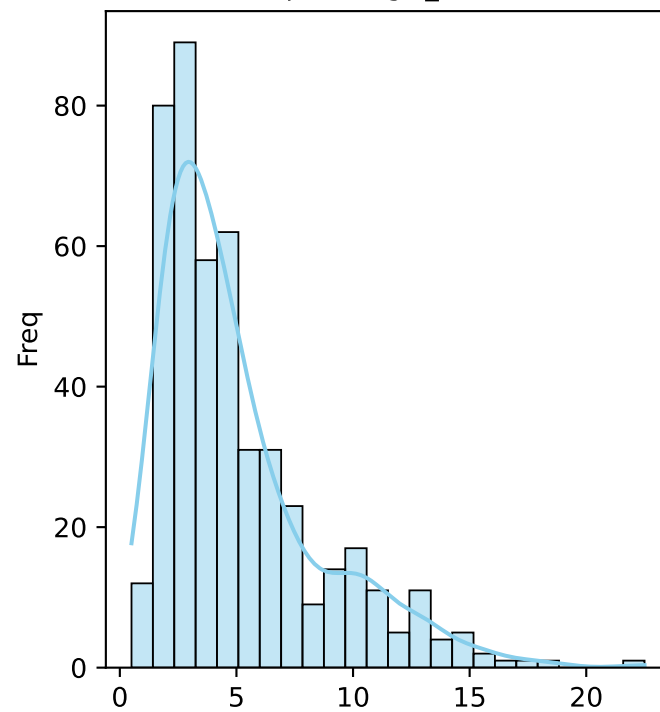
vital_days_after_surgery



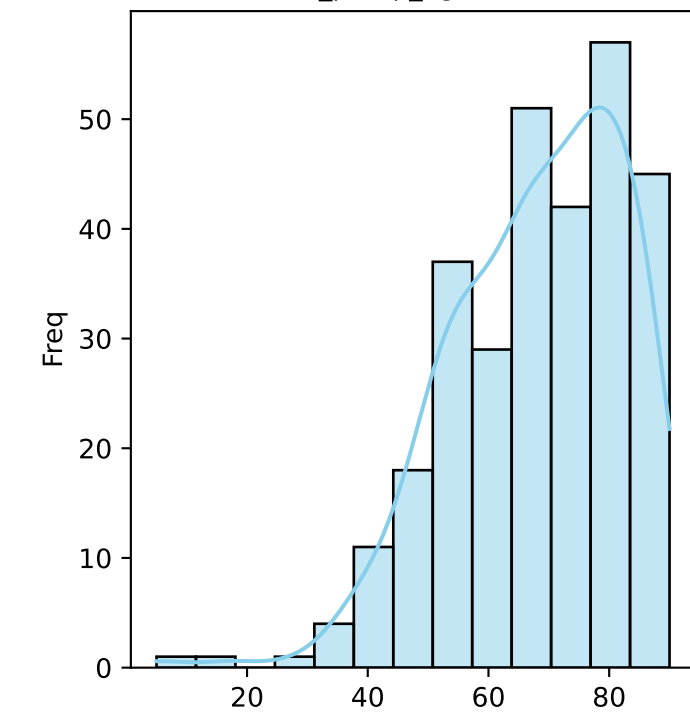
operative_time



pathologic_size

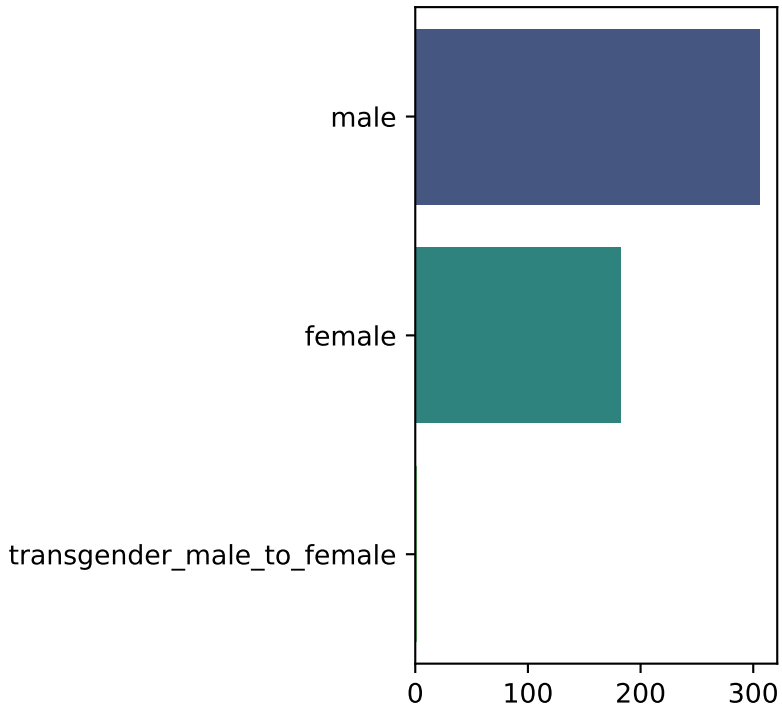


last_preop_egfr.value

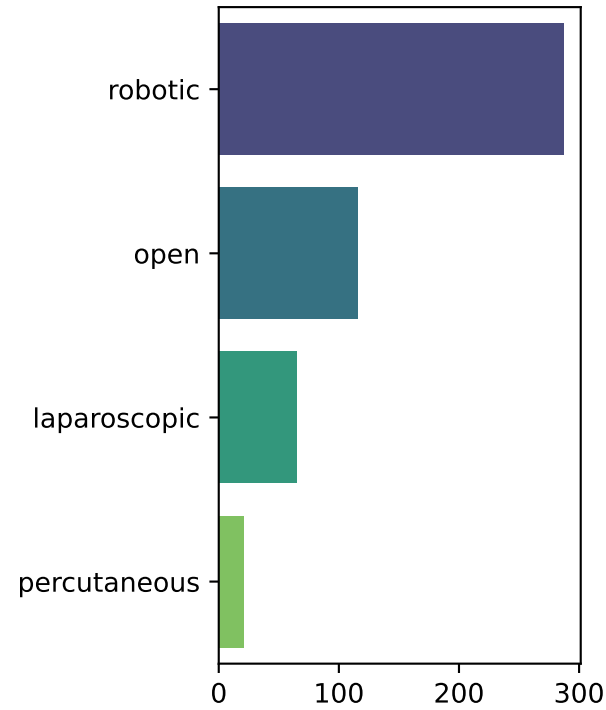


Counts of Key Categorical Variables

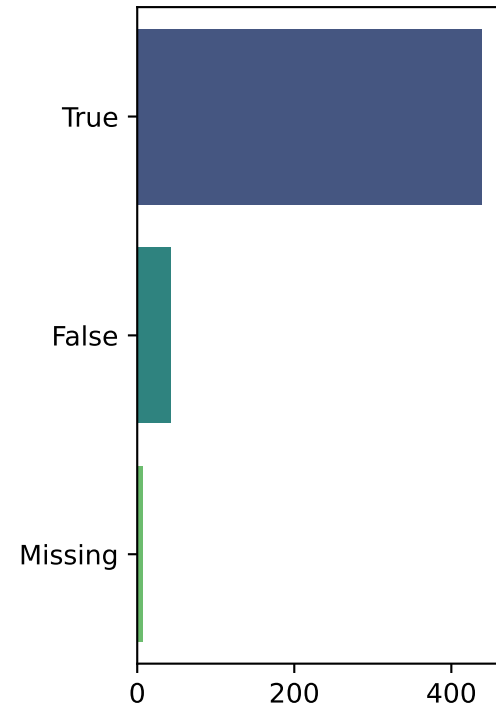
gender



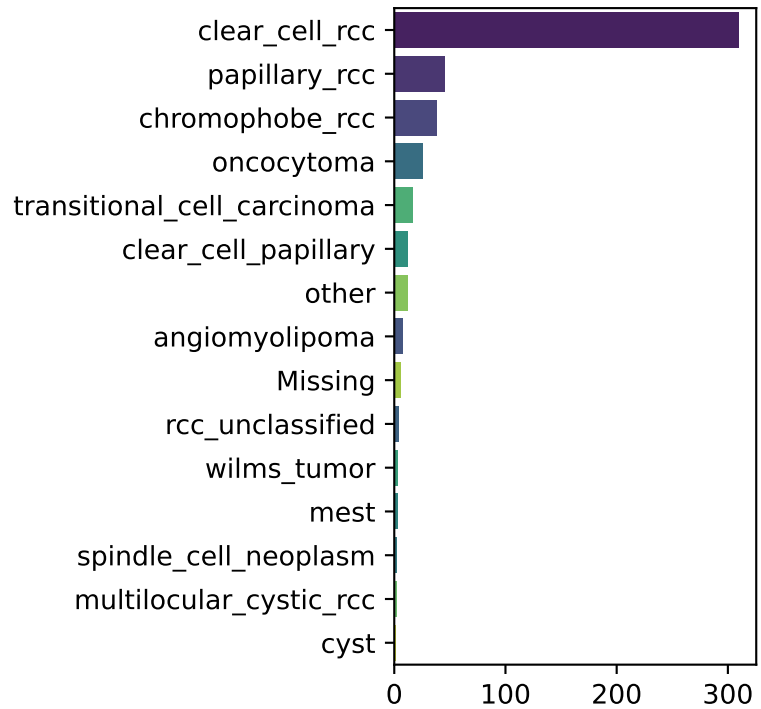
surgery_type



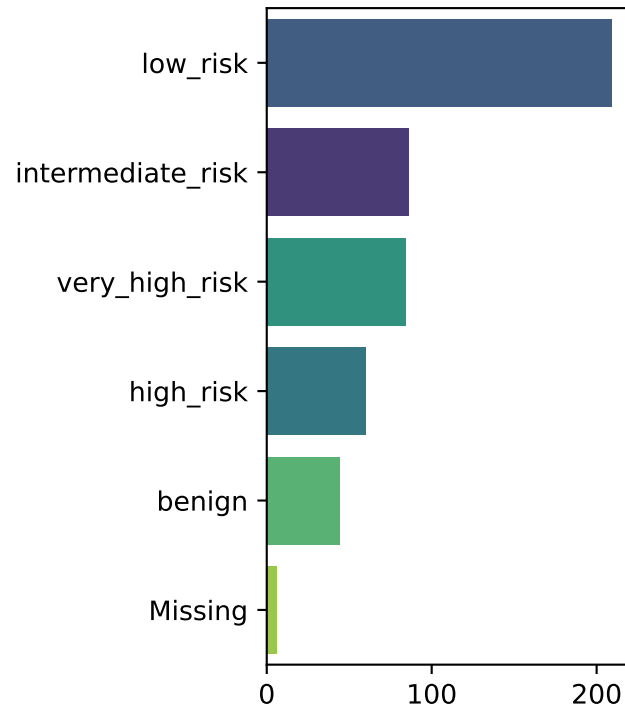
malignant



tumor_histologic_subtype



aua_risk_score



Correlation Heatmap of Numerical Features

